



Operations Research

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Exercise Sheet 4

Demo Exercises

Exercise D4.1. *Dual Problem*

Consider the following linear programs (LPs):

$$\begin{array}{ll}\text{Max} & 4x_1 - 7x_2 + 8x_3 \\ \text{s.t.} & 3x_1 + \quad + 2x_3 \leq 18 \\ & 5x_1 + 7x_2 - x_3 \geq 5 \\ & -6x_1 + 4x_2 + 3x_3 = 30 \\ & x_1, x_2 \geq 0\end{array}$$

$$\begin{array}{ll}\text{Min} & x_1 + 2x_2 - 6x_3 \\ \text{s.t.} & 2x_1 + \quad + 4x_3 = 12 \\ & \quad - x_2 + 5x_3 \geq 10 \\ & -3x_1 + 3x_2 + x_3 \leq 20 \\ & x_1, x_2 \geq 0\end{array}$$

Determine the corresponding dual LPs.

Exercise D4.2. *Primal-Dual*

$$\begin{array}{ll}(P) \text{ Max} & 2x_1 + 3x_2 + x_3 \\ \text{s.t.} & 2x_1 + x_2 + x_3 \leq 4 \\ & 4x_1 + 2x_3 \leq 6 \\ & 5x_2 + 3x_3 \leq 8 \\ & x \in \mathbb{R}_{\geq 0}^3\end{array}$$

$$\begin{array}{ll}(D) \text{ Min} & 4y_1 + 6y_2 + 8y_3 \\ \text{s.t.} & 2y_1 + 4y_2 \geq 2 \\ & y_1 + 5y_3 \geq 3 \\ & y_1 + 2y_2 + 3y_3 \geq 1 \\ & y \in \mathbb{R}_{\geq 0}^3\end{array}$$

- Specify all primal and dual complementary slackness conditions for (P) and (D).
- Let $x^* = (3/2, 1, 0)^T$ be a primal point. Use the complementary slackness conditions to check whether x^* is an optimal solution for (P).
- Is the dual problem (D) unbounded, infeasible or does it have at least one finite optimal solution? Answer this question without solving the dual problem. Justify your answer.

Exercise D4.3. *Transportation Problem*

Consider the following transportation problem: You are responsible for distributing food supplies from collection points $i \in I$ to various food banks $j \in J$. Each collection point $i \in I$ has a supply capacity of food $s_i \geq 0$, and each food bank $j \in J$ has a demand $d_j \geq 0$ for food. Let $c_{ij} \geq 0$ denote the cost of transporting one unit of food from collection point $i \in I$ to food bank $j \in J$.

- a) Formulate an LP (P) to determine a cost-minimizing transportation plan. Use variables $x_{ij} \geq 0$ for all $i \in I, j \in J$, where x_{ij} denotes the amount of food transported from collection point i to food bank j . Please ensure that (1) the demand at each food bank is fully met, and (2) no more food is transported from a collection point than is available.
- b) Formulate the dual problem (D) corresponding to (P).
- c) The dual problem can be interpreted as the optimization problem of an external transport company. What are the meanings of the dual variables in this context?